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Total Thyroidectomy is Associated with Increased Risk of Complications for Low- and High-Volume Surgeons

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ABSTRACT

Background. There has been an increased use of total thyroidectomy (TT), including in the management of benign thyroid diseases. We sought to compare the risk of complications between TT and unilateral thyroidectomy (UT) and to evaluate the effect of surgeon's experience on outcomes.

Methods. Nationwide Inpatient Sample from 2003 to 2009 was used to perform cross-sectional analysis of all adult patients who underwent TT and UT for benign or malignant conditions. Logistic regression was used to evaluate outcomes and to provide correlation between outcome and surgeon volume. Surgeon volume was categorized as low or high (performing <10 or >99 thyroid operations/year, respectively).

Results. A total of 62,722 procedures were included. Most cases were TT (57.9 %) performed for benign disease. There was a significantly increased risk of complication after TT compared to UT (20.4 vs. 10.8 %: $p < 0.0001$). High-volume surgeons performed only 5.0 % of the procedures overall, with 62.6 % of the high-volume surgeon procedures being TTs. Low-volume surgeons were more likely to have postoperative complications after TT compared to high-volume surgeons (odds ratio 1.53, 95 % confidence interval 1.12, 2.11, $p = 0.0083$). Mean charges were significantly higher for TT compared to lobectomy ($19,365 vs. $15,602, $p < 0.0001$), and length of stay was longer for TT compared to lobectomy (1.63 vs. 1.29 days, $p < 0.0001$).

Conclusions. TT is associated with a significantly higher risk of complications compared to UT even among high-volume surgeons. Higher surgeon volume is associated with improved patient outcomes.

Thyroid disorders are some of the most common endocrine diseases in the U.S. population, with prevalence rates ranging from 5 to 20 %.[1] While the management of thyroid cancer typically involves total thyroidectomy (TT), the extent of thyroidectomy for benign conditions and indeterminate nodules is still a matter of much debate.[2–4] Concerns over the potential need for hormone replacement therapy in some cases of unilateral surgery, the possibility of a second surgery if the diagnosis of cancer is made postoperatively, and the potential for benign nodular recurrence in the remaining lobar remnant have led some to argue in favor of TT for benign conditions. This might be a cause for the recently reported increased use of TT for the management of benign thyroid conditions in America over the last 15 years.[5] This report showed a significant increase in postoperative morbidity after TT compared to unilateral thyroidectomy (UT).

Many studies have demonstrated improved outcomes after complex surgeries in high-volume centers.[6–9] Several studies have shown that increased surgeon experience results in better patient outcomes after a variety of operations, including thyroid surgery.[10–23]

Theodore Kocher was the first to demonstrate a significant reduction in operative mortality with greater surgeon experience after thyroid surgery nearly a century ago.[24] When evaluating the trend overall for greater number of TTs, one must appreciate that the majority of thyroid surgery tends to be performed by low-volume surgeons performing fewer than three thyroid operations per year.[25,26] Furthermore, the current literature fails to

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adequately distinguish the effect of surgeon volume on the specific type of thyroid procedure performed. Although a topic of debate, some evidence suggests that there is no significant difference between the risks of complication after UT compared to TT for benign disease.[27–31]

The American Thyroid Association guidelines recommending that thyroid lobectomy alone may be sufficient treatment for isolated indeterminate solitary nodules in those who prefer a more limited surgical procedure, and for those with small (<1 cm), low-risk, unifocal, intrathyroidal papillary carcinomas in the absence of prior head and neck irradiation or radiologically or clinically involved cervical nodal metastases.[32] Despite these recommendations, the increasing trend in America toward TT for management of benign thyroid conditions and for microcarcinomas led us to undertake the present study in an effort to compare UT and TT procedures and to determine whether surgeon experience has an effect on outcomes.

MATERIALS AND METHODS

A cross-sectional analysis was performed using the Nationwide Inpatient Sample (NIS) database for the years 2003–2009. NIS is part of the Healthcare Cost and Utilization Project, sponsored by the Agency for Healthcare Research and Quality. This is the largest all-payer inpatient care database publicly available in the United States. It contains data from approximately 8 million hospital stays from about 1,000 hospitals sampled to approximate a 20 % stratified sample of U.S. community hospitals.[33] International Classification of Disease, 9th revision (ICD-9), was used to define diagnoses and procedures of interest.

The study population consisted of adult (≥18 years) inpatients who underwent TT (ICD-9: 06.4), or UT (ICD-9: 06.2) as the primary procedure. Primary diagnosis was classified into malignant or uncertain behavior neoplasms of the thyroid (ICD-9: 193, 237.4), benign thyroid diseases (ICD-9: 245.0, 245.1, 245.2, 245.3, 245.4, 245.8, 245.9, 240.0, 240.9, 241.0, 241.1, 241.9, 246.0, 246.1, 246.2, 246.3, 246.8, 246.9, 242.10, 242.11, 242.20, 242.21, 242.30, 242.31, 242.80, 242.81, 242.90, 242.91, 226), or Graves disease (ICD-9: 242.0, 242.00, 242.01).

The main study outcomes included postoperative complication, hospital charges, and hospital length of stay (LOS). Dichotomized complications were defined as the presence of one or more of the general or specific complications based on the secondary diagnoses made during the hospital stay. The two independent variables considered were surgeon volume and type of thyroidectomy. Surgeon volume was classified on the basis of the number of thyroidectomies performed per year as low volume (<10 thyroidectomies/year), intermediate volume (10–99 thyroidectomies/year), and high volume (>99 thyroidectomies/year).

Other secondary independent factors that were assessed as potential confounders were: (1) patient demographics: age (<64, 64–79, ≥80 years old), gender, and race (white, African American, Hispanic, other); (2) socioeconomic factors: household income (quartile classification) and main payer of health service (Medicare, Medicaid, private insurance, self-pay); (3) clinical factors: primary diagnosis (malignant, benign, Graves disease), obesity (body mass index ≥30 kg/m^2), inpatient death, whether neck dissection was performed, and a modification of the Charlson comorbidity index score (CCIS) used to assess patient comorbidities (none, mild, moderate, severe); (4) hospital characteristics: hospital location (urban, rural), geographic region (Northeast, Midwest, South, West), hospital bed size (small, medium, large), teaching status (teaching, non-teaching), and hospital volume dichotomized on the basis of the number of thyroidectomies performed per year, such that high-volume hospitals were defined as those above the 75th percentile.[34]

Statistical analysis used weighted data reflecting the national estimate. The recorded weights are available in the NIS data and calculated on the basis of the stratification variables that were used in sampling methodology. These variables are hospital geographic region, urban/rural location, teaching status, bed size, and ownership.

Cross-tabulation and χ^2 test were used to examine the association between each of the independent factors and postoperative complications. Factors with significant association were considered confounders and were included in multivariate logistic regression models. Odds ratio (OR) and 95 % confidence interval (CI) were calculated for the relation between postoperative complication (dependent variable) and surgeon volume and type of thyroidectomy (independent variables). A test for the interaction term between surgeon volume and type of thyroidectomy was performed. In cases of significant results, two types of stratification were followed: the first was by surgeon volume when the model was looking at associations between postoperative complication and type of thyroidectomy; the second was by the type of thyroidectomy when the model was looking at the associations between postoperative complication and surgeon volume. ANOVA was used to test for differences in charges and LOS between surgeon volume and type of thyroidectomy. When ANOVA demonstrated significant findings in the case of surgeon volume, post hoc hypothesis was proposed that examined the difference between charges and LOS for high surgeon volume compared to the other groups. Significance level was set as ($\alpha = 0.05$). All data analyses were performed by SAS 9.2 for Windows (SAS Institute, Cary, NC, USA).

TABLE 1 Descriptive statistics of study population in relation to presence of postoperative complications

Characteristic	Weighted % for			p[a]
	All cases (n = 62,722)	Postoperative complications present (n = 10,257)	Postoperative complications absent (n = 52,465)	
Age (year)				
<65	76.1	72.1	76.9	
65–79	20.7	23.3	20.2	
≥80	3.1	4.6	2.9	<0.0001
Gender				
Male	19.4	18.1	19.7	
Female	80.6	81.9	80.3	0.0009
Race				
White	71.7	71.4	71.7	
Black	11.5	12.0	11.3	
Hispanic	8.5	9.2	8.4	
Other	8.4	7.4	8.6	0.0258
Household income quartile				
<$39,000	20.0	22.0	19.6	
$39,000–$47,999	23.2	23.5	23.1	
$48,000–$62,999	25.1	25.4	25.0	
>$62,999	31.8	29.1	32.3	<0.0001
Service payer				
Medicare	23.7	28.2	22.8	
Medicaid	7.7	7.5	7.7	
Private/HMO	66.4	61.9	67.2	
Self-pay	2.3	2.4	2.3	<0.0001
Surgeon volume				
Low	50.2	55.4	49.2	
Intermediate	44.8	41.1	45.5	
High	5.0	3.5	5.2	0.0001
Thyroidectomy				
Unilateral	42.1	27.7	44.9	
Total/complete	57.9	72.3	55.1	<0.0001
Thyroid diagnosis				
Graves disease	3.3	3.6	3.2	
Benign	60.8	56.4	61.7	
Malignant	35.9	39.9	35.1	<0.0001
Neck dissection				
Yes	0.4	0.7	0.4	
No	99.6	99.3	99.6	0.0007
In-hospital mortality				
Yes	0.0	0.2	0.0	
No	100.0	99.8	100.0	<0.0001
Hospital region				
Northeast	28.6	22.8	29.7	
Midwest	19.1	21.4	18.7	
South	28.1	34.2	26.9	
West	24.2	21.7	24.7	<0.0001
Hospital volume[b]				
Low	74.7	77.4	74.1	

TABLE 1 Continued

Characteristic	Weighted % for			p^a
	All cases (n = 62,722)	Postoperative complications present (n = 10,257)	Postoperative complications absent (n = 52,465)	
High	25.3	22.6	25.9	0.0046
Hospital bed size				
Small	10.8	9.1	11.1	
Medium	21.5	20.9	21.7	
Large	67.7	70.0	67.2	0.0083
Hospital location				
Rural	6.7	7.0	6.6	
Urban	93.3	93.0	93.4	0.4117
Hospital teaching status				
Nonteaching	42.2	43.1	42.1	
Teaching	57.8	56.9	57.9	0.3721
Charlson index				
None	69.2	61.6	70.7	
Mild	25.0	29.4	24.1	
Moderate	4.9	7.5	4.4	
Severe	0.9	1.5	0.7	<0.0001
Obesity (BMI ≥30 kg/m^2)				
Yes	7.3	10.1	6.7	
No	92.7	89.9	93.3	<0.0001

BMI body mass index

[a] χ^2 test

[b] Low-hospital volume indicates <75th percentile of the volume of procedures performed in calendar year; high-hospital volume, >75th percentile

RESULTS

A total of 62,722 discharge records were identified between 2003 and 2009 (Table 1). The mean age of the study population was 51.9 (±0.11) years. The majority of the study population was white (71.7 %) and female (80.6 %), and had private insurance (65.6 %). Most patients did not have other comorbidities at time of admission (68.4 %), and 35 subjects died during their hospital stay. Low-volume surgeons performed 50.2 % of all thyroidectomies, while high-volume surgeons performed only 5 % of thyroidectomies. Slightly more than half of the thyroidectomies were TT (57.9 %), with 53.5 % of the procedures performed by low surgeon volume being TT, while for intermediate and high surgeon volumes it was 61.2 and 62.6 %, respectively. In regard to diagnosis, benign thyroid disease was most common (60.8 %), Graves disease was least common (3.3 %), and malignant conditions made up the rest.

Univariate analysis revealed that a greater overall number of complications occurred after operations performed by low-volume surgeons, 3,492 (55.4 % of total complications), and intermediate-volume surgeons, 2,577 (41.1 %), compared to high-volume surgeons, 211 (3.5 %) ($p = 0.0001$) (Table 1). Most complications (72.3 %) were after TT compared to lobectomy (27.2 %) ($p < 0.0001$). More than half of the complications (56.4 %) occurred after operations for benign thyroid disease compared to malignant conditions (39.9 %). Obesity ($p < 0.0001$) and CCIS ($p < 0.0001$) both demonstrated significant association with the study outcome. Postoperative complications also appeared to be significantly different among variables considered as proxies for socioeconomic status of the patient: household income categories ($p < 0.0001$) and type of insurance ($p < 0.0001$). Hospital characteristics such as volume ($p = 0.0046$), bed size ($p = 0.0046$), and region ($p < 0.0001$) demonstrated significant association with complication; however, hospital teaching status and location in terms of rural or urban based did not.

When performing further analyses, the following variables, besides the main factors of interest, were included in multivariate models as potential confounders: age, gender, race, household income, payer of health service, primary

TABLE 2 Weighted odds ratio of postoperative complication by type of thyroidectomy, overall and stratified by surgeon volume

Surgeon volume	Procedure	Unadjusted			Adjusted[a]		
		uOR	95 % CI	*p*	aOR	95 % CI	*p*
Overall	Unilateral	Reference			Reference		
	Total/complete	2.19	2.042–2.349	<0.0001	2.154	1.987–2.334	<0.0001
Low[b]	Unilateral	Reference			Reference		
	Total/complete	2.368	2.178–2.576	<0.0001	2.368	2.142–2.618	<0.0001
Intermediate[c]	Unilateral	Reference			Reference		
	Total/complete	2.109	1.892–2.351	<0.0001	1.914	1.706–2.148	<0.0001
High[d]	Unilateral	Reference			Reference		
	Total/complete	2.066	1.530–2.789	<0.0001	1.824	1.458–2.281	<0.0001

uOR unadjusted odds ratio, *aOR* adjusted odds ratio, *CI* confidence interval

[a] Adjusted for age, gender, race, household income, type of insurance, primary diagnosis, obesity, inpatient death, neck dissection, Charlson index, hospital geographic region, hospital bed size, and hospital volume

[b] Low surgeon volume is <10 thyroidectomies/year

[c] Intermediate surgeon volume is 10–99 thyroidectomies/year

[d] High surgeon volume is >99 thyroidectomies/year

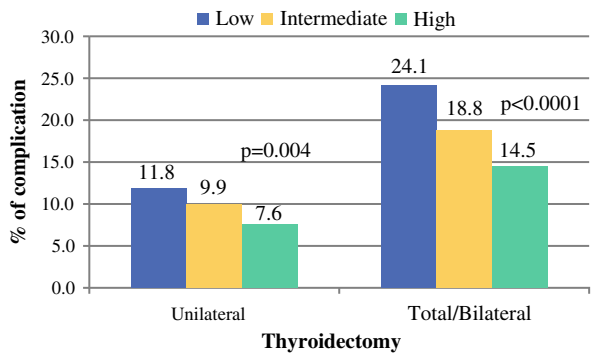

FIG. 1 Risk of complication by surgeon volume and type of thyroidectomy. Low surgeon volume is <10 thyroidectomies per year; intermediate, 10–99 thyroidectomies per year; and high, >99 thyroidectomies per year

diagnosis, obesity, inpatient death, neck dissection, CCIS, hospital geographic region, hospital bed size, and hospital volume.

Postoperative complications were reported in 16.4 % of total cases. Overall risk of complication after TT was 20.4 %, which was significantly higher than UT (10.8 %, $p < 0.0001$). Hypocalcaemia was by far the most common complication after both procedures; however, it was significantly less prevalent after UT (7.1 vs. 16.1 %, $p < 0.0001$). In addition, TT had significantly higher risk for respiratory complication (0.84 vs. 1.34 %, $p < 0.001$), bleeding (0.15 vs. 0.23 %, $p = 0.0403$), hematoma (1.24 vs. 1.54 %, $p = 0.0027$), tracheostomy (0.004 vs. 0.024 %, $p = 0.0493$) and vocal cord paralysis (0.59 vs. 1.33 %, $p < 0.001$) compared to UT. Controlling for confounders in a multivariate model demonstrated that TT was more likely to result in complication compared to UT (OR 2.15, 95 % CI 1.99–2.33, $p < 0.0001$), and that this remained true regardless of surgeon volume (Table 2).

The risk of complication for each of the surgeon volume groups showed an inverse distribution for both types of thyroidectomy procedures (Fig. 1). The risk of complication after TT decreases as the surgeon volume becomes higher (24.1, 18.8, 14.5 %, $p < 0.0001$ for low, intermediate, and high-volume surgeons, respectively), and despite UT having lower risks overall when compared to TT, UT also exhibited a similar relationship with decreasing risk as surgeon volume increased (11.8, 9.9, 7.6 %, $p = 0.004$).

Low-volume surgeons had higher overall risk of complications (18.4 %) compared to high-volume surgeons (11.9 %) ($p < 0.0001$) (Table 3). Examining the risk of each possible complication revealed a consistent decrease in risk with increasing surgeon volume. Furthermore, none of the subjects treated by high surgeon volumes had any cardiovascular, renal, bladder, wound complications or required tracheostomy. After stratification by type of thyroidectomy, however, it appeared that the risk of complication was not related to surgeon volume when performing UT ($p = 0.2648$) (Table 4). However, when performing TT, surgeons with low volume were significantly more likely to encounter complications compared to high-volume surgeons (OR 1.53, 95 % CI 1.12–2.11, $p = 0.0083$). There was no difference in the risk of complications when TT was performed by either intermediate- or high-volume surgeons ($p = 0.2724$).

Total charges incurred after TT were significantly more than UT. This was true when no postoperative complications occurred ($19,365.00 ± 448.06 vs. $15,602.00 ± 283.94, $p < 0.0001$) as well as when there were

TABLE 3 Type of postoperative complication in relation to surgeon volume

Type of complication	Weighted % for[a]				p^c
	All cases (n = 62,722)	Low surgeon volume[b] (n = 18,954)	Intermediate surgeon volume[c] (n = 16,797)	High surgeon volume[d] (n = 1,799)	
One or more	16.371	18.426	15.332	11.921	<0.0001
Cardiovascular	0.060	0.077	0.042	0.000	NA
Pulmonary	1.135	1.639	0.530	0.287	<0.0001
Renal	0.108	0.164	0.049	0.000	NA
Bleeding	0.198	0.274	0.103	0.000	NA
Infection/sepsis	0.085	0.118	0.061	0.061	0.2136
Wound complication	0.002	0.005	0.000	0.000	NA
Shock	0.006	0.016	0.000	0.000	NA
Neck hematoma	1.411	1.727	1.147	0.549	<0.0001
Neck seroma	0.038	0.046	0.011	0.000	NA
Cystitis	0.017	0.020	0.025	0.000	NA
Hoarseness	0.551	0.586	0.586	0.332	0.4183
Tracheomalacia	1.084	1.219	0.973	0.892	0.2246
Tracheostomy	0.016	0.011	0.005	0.000	NA
Hypocalcaemia	12.333	13.665	12.128	9.300	0.0091
Vocal fold paralysis	1.017	1.035	0.790	1.005	0.5227

NA not applicable as a result of small sample size

[a] Column percentage for each type of complication; percentage complements absence of that particular complication

[b] Low surgeon volume is <10 thyroidectomies/year

[c] Intermediate surgeon volume is 10–99 thyroidectomies/year

[d] High surgeon volume is >99 thyroidectomies/year

[e] χ^2 test

TABLE 4 Weighted odds ratio of postoperative complication by the surgeon volume, overall and stratified by type of thyroidectomy

Procedure	Surgeon volume[a]	Unadjusted			Adjusted[a]		
		uOR	95 % CI	p	aOR	95 % CI	p
Overall	Low[b]	1.667	1.173–2.371	0.0044	1.462	1.033–2.070	0.0321
	Intermediate[c]	1.337	0.950–1.882	0.0963	1.205	0.867–1.673	0.2663
	High[d]	Reference			Reference		
Unilateral	Low[b]	1.635	1.030–2.596	0.0372	1.351	0.796–2.292	0.2648
	Intermediate[c]	1.335	0.849–2.099	0.2112	1.306	0.784–2.174	0.3053
	High[d]	Reference			Reference		
Total/bilateral	Low[b]	1.875	1.387–2.533	<0.0001	1.533	1.116–2.105	0.0083
	Intermediate[c]	1.363	1.020–1.821	0.0365	1.182	0.877–1.592	0.2724
	High[d]	Reference			Reference		

uOR unadjusted odds ratio, *aOR* adjusted odds ratio, *CI* confidence interval

[a] Adjusted for age, gender, race, household income, type of insurance, primary diagnosis, obesity, inpatient death, neck dissection, Charlson index, hospital geographic region, hospital bed size, and hospital volume

[b] Low surgeon volume is <10 thyroidectomies/year

[c] Intermediate surgeon volume is 10–99 thyroidectomies/year

[d] High surgeon volume is >99 thyroidectomies/year

complications ($28,295.00 ± 772.35 vs. $ 23,065.00 ± 719.04, $p < 0.0001$). Furthermore, TT resulted in significantly longer LOS regardless of the presence of postoperative complication: absence of complication (1.63 ± 0.02 vs. 1.29 ± 0.01 days, $p < 0.0001$), presence of postoperative complication (3.08 ± 0.07 vs. 2.45 ± 0.08 days, $p < 0.0001$).

When examining the relationship between charges and surgeon volume, post hoc multivariate analysis did not reveal a difference between high surgeon volumes compared to the other two groups ($16,707.00 ± 1,726.15 vs. $16,543.00 ± 413.61, $p = 0.9087$) when there were no complications. When complications occurred, however, total charges after operation by the high-volume group were significantly less than the other two groups ($20,874.00 ± 1,952.54 vs. $25,074.00 ± 731.25, $p = 0.0467$). Both univariate and multivariate analysis showed that the LOS after surgery by high-volume surgeons was significantly shorter both in the presence (1.67 ± 0.2 vs. 2.91 ± 0.07 days, $p < 0.0001$) or absence of complications (1.1 ± 0.1 vs. 1.5 ± 0.1 days, $p < 0.0001$).

DISCUSSION

There has been an increasing trend toward performing TT, especially for the management of benign disease, in the United States over the last 15 years.[5,20] Also, TT is offered for some cases of microcarcinoma and low-grade thyroid malignancies.[35–37] We and others have investigated the relationship between surgeon volume and outcomes.[19–23] However, to our knowledge, none of these studies has looked specifically at different surgeon volume groups to determine the effect of the type of thyroidectomy performed.

Performance of TT for preoperatively defined benign lesions offers the benefit of adequate removal of disease, prevention of recurrence, and avoidance of need for completion surgery when malignancy is diagnosed.[30,38] Arguments against TT in this situation surround concerns that it has potentially higher complication rates than UT. Despite some previous reports, results from our study suggest that TT procedures do have higher associated risks of complications compared to UT (20.4 vs. 10.8 %, respectively), and that this is true even for high-volume surgeons.[27–31] Interestingly, more than half of these complications (56.4 %) occurred after operations for benign disease. We therefore think that TT is a procedure that should be applied judiciously, regardless of surgical volume of the provider. Given our data, the use of TT should be especially cautiously applied when dealing with benign goiter or low-grade malignancies.

Reports have emerged evaluating outcomes after stratification by surgeon volume and experience. Several studies specifically looking at thyroid surgery have provided evidence that complication rates are higher when low-volume surgeons perform TT.[5,19–23,39,40] These results are supported in our findings; overall, low-volume surgeons were significantly more likely to have complications compared to high-volume surgeons (18.4 vs. 11.9 %, respectively) (Table 3). Furthermore, when stratifying by type of operation, low surgeon volume was associated with significantly higher risk of complication after TT compared to high-volume surgeons (24.1 vs. 14.5 %, respectively) (Fig. 1; Table 4).

Aside from complications, charges and LOS were also examined. Both were significantly more after TT, regardless of whether complications occurred. When stratified by surgeon volume, charges were significantly less for high-volume compared to low-volume surgeons only when complications were present ($20,874 vs. $25,074). LOS was significantly shorter after operation by high-volume surgeons whether a complication resulted (1.67 vs. 2.91 days) or not (1.1 vs. 1.5 days). In agreement with Sosa et al., we demonstrate evidence of an inverse relationship between surgeon volume and LOS on a national level that is not predicted by changes in hospital volume.[19]

Our study suggests that the type of thyroid resection does have significant influence on morbidity, and hence, surgeon skills and experience are not the sole determinant of complication rates.[19,27] Rather, there is an inherent increase in risk in the offering of a TT that persists across all surgeon volumes. As such, all surgeons, regardless of experience, should by mindful of the intrinsic risk when performing a TT.

There are several limitations and shortcomings to this type of study that need to be mentioned. The data set does not allow us to fully adjust for the extent of thyroid disease or stage of thyroid cancer. The use of the NIS database underestimates 30-day complication and mortality rates after all procedures. Complications or deaths that occur after hospital discharge are not included in the data set. Furthermore, coding inconsistencies exist, which may reflect the higher-than-anticipated rate of hypocalcemia as a complication, especially after UT.[41] Complications may be diagnosed and followed only on an outpatient basis or coded as separate diagnoses when requiring readmission. Furthermore, because some complications, such as unilateral cord or parathyroid injury, may be underappreciated as a result of the lack of clinically objective findings resulting in failure to code these complications during the index hospitalization, it is possible that the database will underestimate the complication rates after the unilateral procedure subset to a larger extent.

Because we are only capturing the data of patients admitted to the hospital, it is possible that these patients represent a sicker demographic than patients typically undergoing outpatient operations, and thus complications may be more robust among this patient population. Additionally, because the NIS database does not capture ambulatory surgery cases, the LOS will be skewed upward because we are only capturing patients who are admitted.

Unlike some studies that strictly examine complications felt to be directly associated with thyroid surgery, such as recurrent laryngeal nerve injury, hypoparathyroidism, or neck hematoma, this study captured all perioperative complications that occurred before hospital discharge. Because some complications are missed by this database, it is likely that our results represent a conservative estimate of the risks that may occur after thyroid surgery. Furthermore, the NIS database uses weighted data points in an attempt to scale its data to reflect representative information at the national level. All calculations conducted in this study used the provided weighted data to produce the most accurate information on a national level, which makes these findings closely tailored to the U.S. health system.

In conclusion, this study demonstrates that an inherently higher risk of complications is associated with TT compared to UT and that this risk is evident even among experienced surgeons performing high numbers of thyroid cases each year. Higher surgeon volume, however, is associated with improved patient outcomes after TT, fewer total charges when complications do occur, and a decreased LOS. Overall, we have confirmed that the vast majority of thyroidectomies are performed by the low-volume provider. Given our data, surgeons should be mindful of the inherent risk of offering TT for benign disease, microcarcinomas, and low-grade thyroid cancers.

CONFLICT OF INTEREST The authors declare no conflict of interest.

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